Banks, Peer-to-Peer Lending Platforms and the Transmission of Monetary Policy: Loan-Level Evidence from France

Mattia Girotti (Université Paris Dauphine–PSL), Vittoria Iannotta (HEC Paris), Andrea Polo (LUISS, EIEF, CEPR, ECGI)

Extended abstract

Overview. How do non-banks respond to monetary policy, and how does their response differ from banks'? To answer this question, we combine detailed loan-level data for banks and peer-to-peer platforms in France, rate surprises after policy announcements, and webscraped data on investors' types invetsing on platforms shocks. We compute high-frequency monetary policy shocks, and we compare the response of both lenders to monetary policy. Following a surprise that raises short-term rates by one percentage point, platforms raise their lending rates by 0.12 to 0.16 percentage points less than banks lending to comparable firms in the same quarter, and they cut credit by more. The cut in peer-to-peer lending is driven by the banks and credit institutions that invest through the platforms, not from the individual investors. These results are in contrast with predictions of non-bank lenders expanding credit during tightenings, as work on the aggregate non-bank sector reports.

Institutional details on platforms. Platforms differ from banks in their organizational structure and lending policies. They hold no balance sheet: investors fund loans directly and bear the default risk, so the cost-of-funds channel operates only to the extent that investors reallocate their savings. They assess borrowers with automated algorithms on hard and soft information from the application, and they assign a rate from a grid of credit-risk bins that is revised less than once a year. The consequence is visible in the data: within a platform, firms in the same bin pay almost the same rate, while the same firm borrows from a bank and from a platform at rates several percentage points apart. Relationship lending, which blurs the measurement of both channels in bank data, is weak in this setting.

Data. We combine two loan-level registries held by the Banque de France. The first covers 14 peer-to-peer platforms that report their entire business lending book daily, under an agreement that gives them access to firms' balance sheets and credit ratings in exchange. For each loan we observe the issuance date, the amount, the global interest rate including fees and commissions, the purpose, the maturity, the repayment schedule, default events, and, loan by loan, the share funded by individuals, non-financial firms and credit institutions. When the investors' type shares are missing, we recover them by webscraping investors' websites. The second is the bank side: the survey of new credit agreements for pricing, and the credit registry for outstanding stocks. We match both to FIBEN balance sheets and to the Banque de France rating, and we restrict the sample to 2016Q1–2021Q4, when platform reporting is complete and the rating scale is unchanged. The pricing sample holds about 2,200 platform loans and 700,000 bank loans; the quantity sample holds 38 million bank and 19,000 platform firm–lender–quarter records, of which 14,500 are firms borrowing from a bank and a platform at the same time. We identify policy surprises from high-frequency changes in OIS yields around ECB announcements following Altavilla et al. (2019), and we use the Target factor, rescaled so that one unit is a one percentage point increase in the short-term rate, cumulated to the quarter.

Banks pass through rates, but platforms do not. Comparing rates on loans issued to the same firm cluster in the same quarter, the interaction of the shock with a platform indicator is -0.16 and stays between -0.16 and -0.12 as we tighten the fixed effects from quarter to region $\times$ size $\times$ sector $\times$ rating $\times$ quarter, on 643,000 loans. The result follows from how platforms price: a grid set by the platform and revised infrequently cannot track market rates, whatever the investors who fund the loan would require. The main concern is that platform and

bank borrowers differ, so we match them on age, tangibility, operating margin, financial debt over equity, leverage over profit and assets, exactly within quarter, region, sector, size, maturity and loan type. On the 874 matched observations the rate interaction is -0.31 with lender and matched-firm fixed effects ($p = 0.05$), larger than in the full sample.

Platforms cut quantities more than banks. On the credit registry, we estimate the effects of a rate shock on quantities issued. We measure the quarterly growth of credit to a firm, accounting also for new credit for first-time borrowers using the Davis and Haltiwanger (1992) mesure. The policy-platform interaction is -0.019 with firm $\times$ quarter and firm $\times$ lender fixed effects, and between -0.022 and -0.017 across specifications, on 114,000 firm–lender–quarter observations. Platforms cut credit more than banks after a tightening surprise. Demand is held fixed by the firm $\times$ quarter effects, so the comparison is between a bank and a platform lending to the same firm in the same quarter.

Banks investing through platforms drive the contraction. We observe each investor's share of each loan funded via a platform, so we can ask whether different investors respond heterogeneously to rate shocks. Individuals are the main investor in 83 percent of loans; credit institutions, including banks, hold larger shares of larger loans. Within a platform and a firm $\times$ quarter cell, credit growth funded by banks falls by 0.042 to 0.057 more than credit growth funded by individuals after the same shock. The finding is in contrast with the deposit outflow channel that explains non-bank expansion: individual investors do not replace what the banks withdraw, so platform credit falls.

Contribution. We contribute to the literature on monetary policy transmission by providing granular evidence on the role of non-bank financial intermediaries. The literature asks how non-banks change monetary transmission and reaches opposite conclusions from aggregated categories: Elliott et al. (2022) and Ćučić and Gorea (2023) find that non-banks expand credit after a tightening, while Di Maggio and Kacperczyk (2017) find excess risk-taking at the zero lower bound. We can focus on a specific intermediary, to which we can attribute the finding, and we show that heterogeneity within the non-bank sector reverses the sign. The second studies fintech lending with granular data, so far from one platform at a time (Huang, Li and Wang 2021; Zhu and Lu 2021). We observe the universe of business loans of 14 platforms as a registry, with the investor composition of each loan, which is what lets us trace the response to the funding side.

Work in progress. Two challenges remain open. First, platform borrowers are relatively small, and a fraction of them is not in the Banque de France registries. To ensure we have as many controls as possible: we are recovering firm demographics from INSEE administrative records that the credit registries do not carry. Second, whether platform and bank borrowers are comparable in terms of their risk profile is a question in its own right: the concern is that different lenders attract different borrowers, and that the observed difference in response to monetary policy is due to that selection. We are addressing it by extending the matching and by constructing firm-level distress events – liquidation and judicial restructuring – from the French administrative source BODACC, which also lets us ask whether the two lenders' borrowers fail at different rates.

Keywords: peer-to-peer lending, fintech, monetary policy transmission, credit registry, non-bank finance. **JEL:** E52, G21, G23, E44.